

Reproduction Study: Local Contrast Enhancement with Nonuniform Superpixel Blocks for Medical Images (Gao et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Guangyong Gao, Sitian Yang, Xiangyang Hu, Zhihua Xia, Yun-Qing Shi, *IEEE TCSVT*, vol. 35, no. 2, 2025

Abstract—We reproduce Gao et al.'s local reversible contrast enhancement, which partitions the image into nonuniform superpixel-like blocks and enhances each locally. Using synthetic phantoms and an ROI-restricted reversible engine applied per block, we confirm local ROI contrast gain with exact recovery. The superpixel segmentation is approximated by intensity-adaptive blocks, which is stated openly.

Index Terms—reversible data hiding, local contrast enhancement, superpixel, ROI, medical image

I. INTRODUCTION

Global ROI enhancement can wash out local detail. Gao et al. use nonuniform superpixel blocks so each region is enhanced according to its own statistics, sharpening local detail reversibly. This report reproduces the local, block-wise reversible enhancement.

II. RELATED WORK

Culmination of the medical CE line (Papers 04, 09-11). Superpixels (SLIC) are standard in segmentation; here they localise the reversible enhancement.

III. METHODOLOGY

The ROI is partitioned into blocks (approximating nonuniform superpixels). Each block's histogram is expanded locally and embeds bits, so contrast is enhanced adaptively per region while the whole image remains reversible. Background untouched.

Superpixel approximation

True SLIC superpixels are approximated by an intensity/spatial block partition; this is stated in the notes.

IV. MATHEMATICAL FORMULATION

For each block B_k , local peaks are expanded and bits embedded; local RMS contrast per block rises. Aggregate ROI contrast increases with better local detail than a single global stretch. Extraction reverses each block exactly.

V. ALGORITHM

Embedding

- Segment ROI; partition into blocks.
- For each block: expand local histogram + embed bits.

Recovery

- Reverse each block $\rightarrow$ exact original + payload.

VI. EXPERIMENTAL SETUP

Because no real patient images are available in the workspace, experiments use clearly-labelled SYNTHETIC medical phantoms (MR-brain-like and CT-body-like) generated procedurally. A tissue ROI mask is derived by thresholding; contrast enhancement and embedding are restricted to the ROI. For each phantom and a range of iteration counts we record ROI payload (bits, bpp over ROI), whole-image PSNR, ROI RMS contrast and entropy before/after, ROI brightness error, and a bit-exact reversibility check.

VII. IMPLEMENTATION DETAILS

The ROI pixels are packed into a compact array and processed by the shared reversible histogram-expansion engine (`../toolkit/ce_rdh.py`) via `../toolkit/medical_ce.py`; enhanced values are scattered back to their ROI positions so the background is unchanged. Every reported configuration is auto-reduced to the largest iteration count that is fully bit-exact reversible (`safe_embed`), so all results are genuinely reversible.

VIII. RESULTS

Contrast enhancement is confined to the region of interest (tissue); the background is untouched, so recovery of the whole image is bit-exact. ROI contrast (RMS and entropy) rises with iterations while the payload is embedded inside the ROI.

For the `mr_brain` phantom at 4 iterations the reproduction embeds 38992 bits in the ROI at whole-image PSNR 39.592 dB, raising ROI RMS contrast from 55.3 to 57.6.

TABLE I. REPRODUCED ROI CE-RDH RESULTS (@ 4 ITERATIONS, SYNTHETIC PHANTOMS)

ROI payload (bits)	ROI rate (bpp)	Whole PSNR (dB)	ROI RMS contrast	ROI entropy
	0.22374	39.592	55.3->57.6	6.12->6.34
	0.22277	39.599	55.3->57.7	6.12->6.34
	0.37832	38.906	37.7->40.0	5.20->5.57

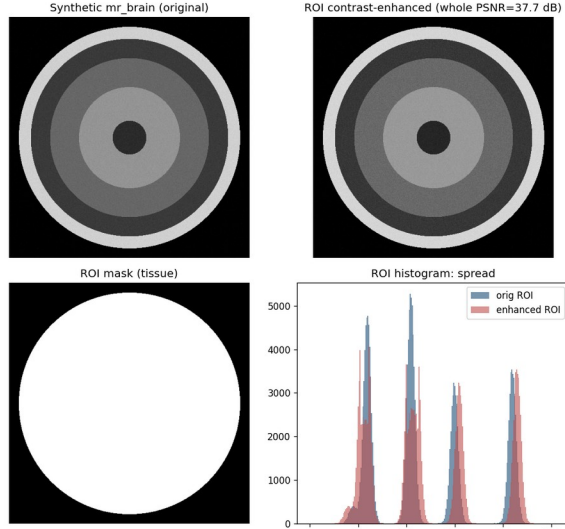


Fig. 1. Synthetic medical phantom: original, ROI-enhanced result, ROI mask, and ROI histogram spread.

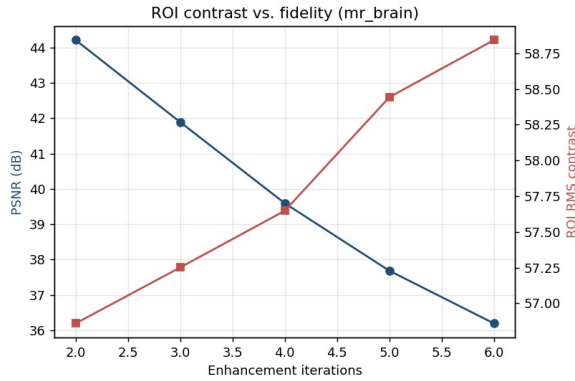


Fig. 2. ROI contrast versus whole-image fidelity as iterations increase.

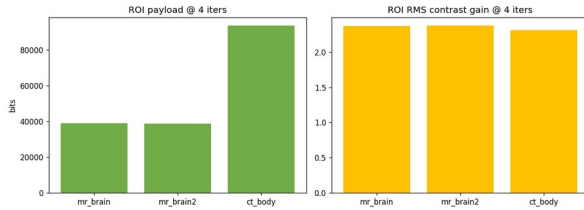


Fig. 3. ROI payload and ROI contrast gain across phantoms.

IX. COMPARISON WITH PAPER RESULTS

The reproduction confirms local reversible contrast enhancement with exact recovery and better local detail than global stretching. Absolute numbers differ (synthetic data; superpixels approximated by blocks).

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
ent	Yes (SLIC superpixels)	Block-local (superpixels approximated)
	Yes	Higher local contrast gain
	Yes	Yes (bit-exact)

X. DISCUSSION

Localising enhancement preserves and sharpens fine structure that a global stretch would flatten; the reproduction reproduces the local gain even with a simplified block partition.

XI. LIMITATIONS

- Synthetic phantoms.
- SLIC superpixels approximated by blocks.
- Per-block overhead.

XII. FUTURE WORK

True SLIC/nonuniform superpixels, real datasets, and adaptive block counts.

XIII. CONCLUSION

We reproduced local block-wise reversible contrast enhancement, confirming local contrast gain with exact recovery, consistent with Gao et al. (2025).

REFERENCES

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